

The Zero-Sum Game of Mutual Fund Marketing

Haiwei Chen
SUF

Ilias Filippou
FSU

Jeong Ho (John) Kim
FSU

Jiangyuan Li
SUF

Dishui Lake Finance Conference International
May 2026

Mutual Fund Marketing: What We Know

- Asset management has grown dramatically — AUM and fund counts have surged worldwide; **investor search costs** are a first-order friction in capital allocation (Hortaçsu & Syverson, 2004; Huang, Wei & Yan, 2007)
- U.S. mutual funds spend roughly **one-third of fee revenue on marketing**; advertising, broker distribution, and platform visibility drive significant fund-level inflows (Sirri & Tufano, 1998; Gallaher, Kaniel & Starks, 2008)
- A growing literature studies **social media marketing** of funds: De Haan et al. (2025), Kacperczyk & Pagnotta (2026), and Li et al. (2026) examine mutual fund livestreaming on retail platforms in China

An open question

Most prior work evaluates marketing at the **fund level**, even though decisions are made and success should be judged at the **family level**. Does fund-level marketing translate into family-level growth?

Mutual Fund Marketing in Practice

(1) Family Brand — TV

Vanguard launches new national TV campaign through AML

By Mark Johnson - 05/01/2022



Vanguard, one of the world's leading investment firms with more than 30 million investors worldwide, has launched a major national advertising campaign through London-based creative agency AML Group.

(2) Star Manager — Print

"Don, convert to a Fidelity Roth IRA in '98 and you can enjoy the luxury of an additional tax benefit." — Peter Lynch

December 31, 1998 is the deadline to take advantage of this limited opportunity.

If you're thinking about converting to a Fidelity Roth IRA, this is the time to do it. Not only can you convert your money today, but you can also enjoy the tax benefits of a Roth IRA. And, by converting before the end of 1998, you can also enjoy the tax benefits of a Roth IRA. And, by converting before the end of 1998, you can also enjoy the tax benefits of a Roth IRA.

Help you make this conversion or simply, "This year, convert your money to a Roth IRA. It's the best time to do it. The tax benefits are great. And, by converting before the end of 1998, you can also enjoy the tax benefits of a Roth IRA. And, by converting before the end of 1998, you can also enjoy the tax benefits of a Roth IRA."

Fidelity Investments

(3) Star Manager — Digital

**“绩”往开来
“优”中选优**

实力老将
刘彦春 匠心掌舵

新兴成长债精选混合基金 冠军

业绩排名	成立时间	基金经理	基金经理
业绩排名	成立时间	基金经理	基金经理
业绩排名	成立时间	基金经理	基金经理
业绩排名	成立时间	基金经理	基金经理

景顺长城绩优成长混合型证券投资基金
2019.6.21-2019.6.28火热发行中 | 募集总额30亿

基金代码: 002412

Vanguard national TV campaign (2022): “I Value Feeling Valued.” Pure brand advertising — no specific fund featured. Builds institutional identity across 30M+ investors.

Fidelity print ad (1998): Peter Lynch endorses Fidelity Roth IRA conversion. Family credibility via star manager — promotes a specific service, not a fund.

(Invesco China) digital ad: star manager (Yanchun Liu) promotes a specific fund with 3-year return data. **Product-level**: direct conversion focus.

Marketing ranges from family-level brand-building (1, 2) to product-level performance showcasing 3/25

This Paper

- We study **social media livestreaming** as a marketing tool used by Chinese mutual fund families on the Tiantian Fund platform, 2020–2025
- We evaluate marketing effects at both the **fund level** and the **family level**, using **25,685** livestreaming events across **180** families and **5,809** funds
- We analyze **795 transcripts** with a large language model to understand what communication strategies attract outside capital
- We examine **competitive dynamics**: do families livestream to grow, or to defend existing market share?

Setting

China's Tiantian Fund platform provides a uniform observation window across all major families, with verified treatment status ("featured" funds in the shopping cart) and precise broadcast timestamps.

Main Findings

- (1) Zero at the family level.** Livestreaming generates large fund-level inflows (+14 pp) but **zero** family-level flow or revenue change. Featured funds gain at the direct expense of non-featured siblings.
- (2) Intra-family capital optimization.** Families feature recently outperforming (undercapitalized) funds. Post-event performance converges — consistent with Berk-Green rational learning. Livestreaming acts as an **internal open market operation**.
- (3) Defensive arms race.** When rivals increase livestreaming by 1 SD, non-livestreamers lose ≈ 27 pp in flows. Families livestream to **defend market share**, not to expand it — a prisoner's dilemma.
- (4) Hard information wins.** Families that do attract outside capital emphasize **past performance** and **social proof** — not emotional appeals or engagement style.

The Rise of Livestream Marketing

How it began (2020):

- COVID-19 disrupted traditional sales channels: in-branch consultations, roadshows, and fund fairs were suspended
- Fund families pivoted to **real-time video broadcasts** on retail platforms to reach investors directly
- By 2021, every major fund family operated its own livestream channel

Regulatory framework (AMAC, 2021):

- Asset Management Association of China formalized guidance on livestream marketing
- Requirements: investor education, product disclosure, compliance pre-approval, and risk warnings

A new marketing ecosystem:

- Third-party platforms (Tiantian, Alipay, JD Finance) host dedicated livestream portals for all families
- Fund families maintain official channels with **verified broadcaster** status
- Broadcasts are scheduled, indexed, and archived — providing a complete observational record

Key shift

From *intermediary-driven* distribution (brokers, bank advisors) to *direct-to-investor* digital marketing — families now communicate directly

Scale of Livestream Marketing

Scale of the phenomenon:

- **25,685** livestreaming events in our sample
- July 2020 – June 2025 (20 quarters)
- Families broadcast **multiple times per month**
- Typical broadcast: 1–2 hours; live audience ranges from hundreds to tens of thousands of viewers

Our platform (Tiantian Fund):

- One of China's top three retail fund platforms
- **5,809** actively managed equity/hybrid funds across **180** fund families

Hosts livestream channels for all major

Adoption over time:

- Near-zero livestream activity before July 2020
- Rapid diffusion: most families adopted within 2–3 quarters of the first movers
- By 2021–2022, livestreaming was universal among active equity fund families
- Adoption is **staggered**: families enter at different quarters – enables DiD identification

Our sample

90,850 fund-quarter observations; unbalanced panel 2020 Q3–2025 Q2. SEs

Structure of a Livestream Event

Key institutional feature:

- Families select **fewer than 4 funds** per broadcast from their portfolio of 25+ products
- The selected funds appear in the **shopping cart** (“Product Basket”) embedded in the livestream interface
- This creates sharp **within-family variation** in promotional intensity — ideal for identification

Pre-planned, not spontaneous:

- Compliance approval is obtained **days in advance**
- Topic, featured funds, and risk disclosures are all fixed before the

Treatment definition

A fund is “**featured**” in quarter q if any share class appears in the shopping cart during any broadcast that quarter.

A fund is “**non-featured**” if its family broadcasts but the fund is not selected.

Within-family, within-quarter comparison.

Identification

Featured and non-featured funds belong to the *same* family in the *same* quarter — family $\times$ quarter FE absorbs all common shocks. The only difference is **which fund the family chose to promote**.

How a Livestream Works

During the broadcast:

- A **fund manager or professional host** presents to live viewers
- Covers market outlook and highlights 1-4 featured funds
- Risk disclosure banners appear per regulatory requirements
- Viewers can ask questions in real-time via the comment interface

The shopping cart ("Product Basket"):

- Icon in the broadcast interface expands to show featured funds
- Displays: fund name, asset class, risk rating, past performance, subscription window

Who presents?

- **Fund managers:** provide technical commentary and portfolio outlook
- **Professional hosts:** deliver polished presentations; common at large families
- **Guest speakers:** economists, market analysts, platform representatives

Identification

Within the same family and the same quarter, some funds are featured while others are not — giving us a clean within-family comparison.

The Tiantian Platform: Interface Walkthrough

(1) Platform Portal



(2) Livestream Room



(3) Shopping Cart



Sample

Livestream data:

- **25,685** events, July 2020 – June 2025
- Source: Tiantian Fund platform
- Records: broadcast date, duration, host identity, featured funds (shopping cart contents)

Fund universe:

- **5,809** actively managed equity and hybrid funds
- **180** fund families
- **90,850** fund-quarter observations
- Excluded: index, closed-end, QDII, and umbrella funds

Descriptive highlights:

- Mean quarterly fund flow: 0.38%, SD: 42%
- **14%** of fund-quarters are “featured”
- Featured funds outperform non-featured siblings in the months before the event

Fund flow data:

- Total Net Assets (TNA) from Wind Financial
- Quarterly returns from fund prospectuses
- Fund characteristics: inception date, manager, benchmark, share class structure

Key Variables: Flow Measures

Fund flow (return-adjusted):

$$\text{Flow}_{i,q} = \frac{\text{TNA}_{i,q} - \text{TNA}_{i,q-1}(1 + r_{i,q})}{\text{TNA}_{i,q-1}}$$

Family flow (AUM-weighted aggregate):

$$\text{FamilyFlow}_{f,q} = \frac{\sum_{i \in f} \Delta \text{TNA}_{i,q}^{\text{adj}}}{\sum_{i \in f} \text{TNA}_{i,q-1}}$$

where $\Delta \text{TNA}_{i,q}^{\text{adj}} = \text{TNA}_{i,q} - \text{TNA}_{i,q-1}(1 + r_{i,q})$

Why return-adjust?

Raw TNA changes mix investor flows with capital gains. Stripping out returns isolates **investor decisions** – new subscriptions minus redemptions.

Why aggregate to the family?

A fund family's total AUM is the sum of its funds' TNAs. If featured funds gain and non-featured lose by the *same* amount, **FamilyFlow = 0** – even though every fund-level regression shows a positive coefficient.

Our main test

Is $\hat{\beta}$ in the family-level regression

Key Variables: Treatment Indicators

Three treatment indicators:

- **Fund Livestream** $_{i,q}$: fund i appears in the shopping cart during any broadcast in quarter q (“**featured**”)
- **Family Livestream (Oth)** $_{i,q}$: family f broadcasts in quarter q but fund i is *not* featured — the **sibling treatment**; captures cannibalization of non-featured funds
- **Family Livestream** $_{f,q}$: family f holds *any* broadcast in quarter q — the **family-level indicator** used in the aggregate test

Descriptive statistics:

Statistic	Value
Fund-quarters featured	1
Fund-quarters with sibling featured	3
Family-quarters with any broadcast	5
Avg. broadcasts per family per quarter	4
Avg. funds per broadcast	2

Within-family variation

Featured and non-featured funds are in the *same* family, *same* quarter — family FE absorbs all common shocks.

Empirical Design

Family-level – does the family grow?

$$\text{FamilyFlow}_{f,q} = \alpha_f + \delta_q + \beta_1 \text{Livestream}_{f,q} + \mathbf{X}'_{f,q-1}\Gamma + \varepsilon_{f,q}$$

Fund-level – who gains, who loses?

$$\text{FundFlow}_{i,q} = \alpha_i + \delta_q + \beta_1 \underbrace{\text{FundLivestream}_{i,q}}_{\text{fund } i \text{ featured}} + \beta_2 \underbrace{\text{FamilyLivestream(Oth)}_{i,q}}_{\text{sibling featured}} + \mathbf{X}'_{i,q-1}\Gamma + \varepsilon_{i,q}$$

Fixed effects:

- Fund / family FE absorb time-invariant unobservables
- Year-quarter FE control for aggregate market conditions
- SE clustered at the family level

Robustness:

- Family $\times$ quarter FE: absorbs all time-varying family shocks; identifies within-family, within-quarter differential

Family-Level Flows: Null Across All Specifications

Eight family-level TWFE specifications:

Variable	Coef.	(SE)
Family First Livestream	3.77	(5.12)
Family Post Livestream	-0.66	(2.34)
Family Livestream	0.90	(2.17)
Family Market Share	-15.67	(18.43)
Market Livestream (Oth)	-0.90	(1.65)
Family Livestream Count	-0.09	(0.50)

Family FE, quarter FE, controls. None significant.

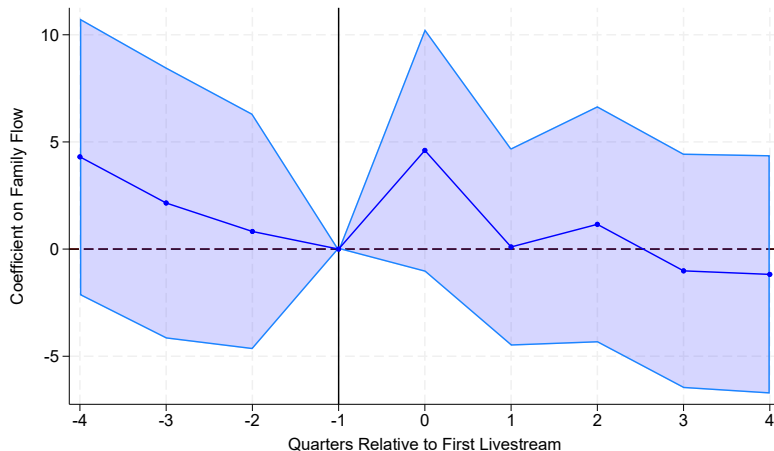
Our finding

Livestreaming generates **zero** change in family-level flows. Confidence intervals rule out economically meaningful effects.

What is happening?

Inflows to featured funds must come from *within* the family – at the expense of non-featured siblings.

Event Study: No Family-Level Response Around Adoption



TWFE estimates of family flows around initial livestream adoption (quarter 0). Pre-event coefficients are flat (parallel trends hold). No sustained increase post-adoption.

Fund-Level Decomposition: The Reallocation Mechanism

	(1)	(2)	(3)	(4)	(5)
Fund First Livestream	14.13*** (1.89)			12.95*** (1.96)	
Fund Livestream			14.87*** (1.85)		13.27*** (1.93)
Family Livestream (Oth)				-1.49** (0.70)	-1.94** (0.88)
Controls	No	No	No	No	Yes

Fund FE, quarter FE. SE clustered at family. ***, ** = 1%, 5%.

Featured funds

Gain **+13 to +15 pp** in quarterly flows — $\approx$ one-third of within-fund flow SD.

Non-featured siblings

Lose **-2 pp** — opposing sign confirms within-family cannibalization.

Robustness: Family $\times$ Quarter Fixed Effects

Replace fund + quarter FE with family $\times$ quarter FE:

- Absorbs *all* time-varying family characteristics: aggregate family flow, performance, market conditions
- Since family aggregate flow = 0 (previous results), this specification mechanically forces the coefficient to equal the **full reallocation magnitude**

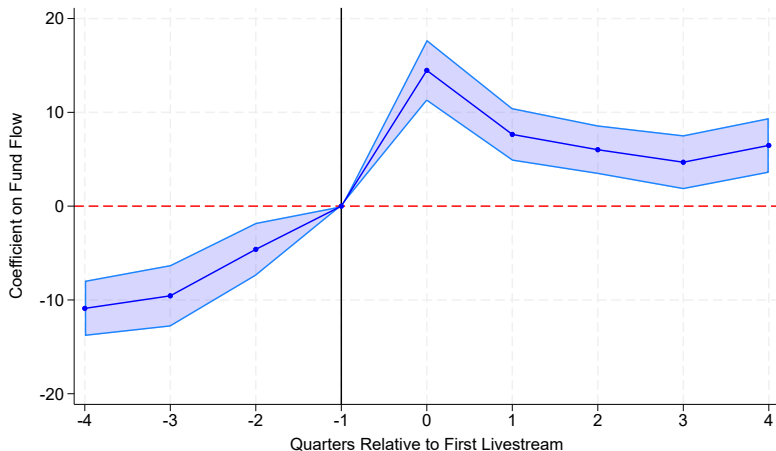
	Fund + Quarter FE	Family $\times$ Quarter FE
Fund Livestream	14.87*** (1.85)	17.59*** (2.25)

SE clustered at family level.

Interpretation

The **+17.6 pp** represents dollar-for-dollar reallocation: every dollar flowing into featured funds is offset by an equal dollar flowing out of non-featured siblings within the same family.

Performance of Featured vs. Non-Featured Funds



Monthly raw returns of featured funds (blue) vs. non-featured siblings relative to the livestream event month. Fund and month fixed effects. Shaded bands: 95% CI.

Interpretation: Families as Open Market Operators

Before the livestream:

- Featured funds **significantly outperform** non-featured siblings in months leading up to the event
- Non-featured funds underperform $\Rightarrow$ they are *overcapitalized* relative to their skill (Berk & Green, 2004)

After the livestream:

- Capital inflows erode featured funds' outperformance
- Capital outflows improve non-featured funds' performance
- Performance gap **compresses** — consistent with flow-performance equilibrium convergence

Mechanism (Berk, 2017)

Families have **superior information** about which funds are undercapitalized. Livestreaming steers existing investors from overcapitalized to undercapitalized funds. **Benefit:** retains investors who might otherwise search outside the family.

Fee Structure and Revenue: Also Zero

Could livestreaming shift flows toward higher-fee products?

- **Management fees** (majority of revenue): **no response** across all specifications and horizons
- **Service fees**: modest temporary increase of a few basis points, then reverts
- **Loads**: slight subsequent reduction
- Net effect: no systematic shift in family's fee structure

Flow-induced revenue (fee $\times$ flow effects combined):

- Statistically and economically insignificant across all specifications and horizons

Bottom line

Families extract **no additional revenue** from livestreaming.

The zero-sum capital reallocation does not improve profitability.

So why livestream at all?

Two answers

1. Intra-family: optimize fund sizing (open market operation)
2. Inter-family: defend market share from rivals

Communication Strategies: What Attracts Outside Capital?

Approach:

- Define *abnormal flows* = residuals from family flow regression on livestream intensity + controls
- Split into High (top 5%) vs. Low (bottom 5%) within each quarterly cross-section
- Sample 5 transcripts per family $\Rightarrow$ **795 transcripts**

LLM scoring (Qwen-2.5-32B):

- **16 dimensions**: sentiment, emotional intensity, sales pressure, past performance, social proof, educational content, risk transparency, ...
- Each scored 1–10 using a *fixed rubric ex ante*

Results: only 2 of 16 dimensions differ

Dimension	Diff.	<i>p</i>
Past Performance Focus	+0.28	.02
Social Proof Reliance	+0.23	.04
Sentiment Positivity	+0.08	.51
Emotional Intensity	−0.05	.68
Sales Pressure	+0.11	.39
Educational Content	+0.03	.82
10 further dims: all insignificant		

Investors respond to **verifiable “hard” information** — past returns and third-party validation — not emotional appeals.

Competitive Equilibrium: The Defensive Motive

Within families (own vs. sibling effects):

- Own-fund livestreaming: +7.2 pp flows
- Sibling livestreaming: -4.9 pp flows
- Net family effect: ≈ 0

Across families (market competition):

- When rivals increase livestreaming by 1 SD, non-livestreamers lose ≈ 27 pp in flows
- Market share in livestreaming activity strongly predicts fund-level flows (+620 pp per unit share)

Implication:

Families that stop livestreaming while rivals continue face *immediate* market share losses.

Prisoner's dilemma

Families must livestream **not to attract new investors** but to **defend existing assets** from more actively marketed alternatives.

Unilateral exit $\Rightarrow$ flow losses.

Mutual participation $\Rightarrow$ zero net gain.

Industry is trapped in a wasteful promotional arms race (Roussanov, Russ & Wei, 2021).

Summary

- (1) Zero-sum at the family level.** Livestreaming generates large fund-level inflows (+14 pp) but zero family-level flow or revenue change. Featured funds gain at the direct expense of non-featured siblings.
- (2) Intra-family capital optimization.** Families feature recently outperforming (undercapitalized) funds. Post-event performance converges — consistent with Berk-Green rational learning. Livestreaming functions as an internal open market operation.
- (3) Defensive arms race.** Families livestream to defend market share, not to expand it. Market-wide competition creates a prisoner's dilemma.
- (4) Hard information wins.** Families that do attract outside capital emphasize past performance and social proof — not emotional appeals or engagement style.

Broader Implications

- **For the mutual fund literature:**

Fund-level marketing effects do not imply family-level growth. Flow-performance sensitivities and marketing effectiveness must be evaluated at the family level.

- **For investor welfare:**

Livestreaming facilitates within-family learning and capital allocation — a genuine benefit. But aggregate marketing may be socially excessive: the industry is in a promotional arms race with no net new capital formation.

- **For policy:**

AMAC's 2021 guidance (investor education, compliance controls) is well-targeted. Our evidence suggests the informative role of livestreaming (hard-info channel) coexists with a defensive market-share motive.

Thank you. Questions welcome.

Appendix: Full Family-Level Flow Regressions

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Family First Livestream	3.77 (5.12)							
Family Post Livestream		-0.66 (2.34)						
Family Livestream			0.90 (2.17)				1.07 (2.19)	
Family Market Share				-15.67 (18.43)				
Market Livestream (Oth)					-0.90 (1.65)			
Family Livestream Count						-0.09 (0.50)		0.01 (0.50)
Market-Family Count							24.55 (20.33)	14.52 (18.47)
Family FE	Y	Y	Y	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y	Y	Y	Y

Appendix: Fund-Level Decomposition (Full)

	(1)	(2)	(3)	(4)	(5)
Fund First Livestream	14.13*** (1.89)			12.95*** (1.96)	
Fund Post Livestream		14.64*** (2.01)			
Fund Livestream			14.87*** (1.85)		13.27*** (1.93)
Family Livestream (Oth)				-1.49** (0.70)	-1.94** (0.88)
Fund FE	Y	Y	Y	Y	Y
Quarter FE	Y	Y	Y	Y	Y
Controls	No	No	No	No	Yes

***, **, * = 1%, 5%, 10%. SE clustered at family level.

Appendix: Share Class Analysis

Specification: Fund-quarter FE absorb all fund-level variation.

$$\text{ClassFlow}_{c,q} = \delta_{f(c),q} + \beta_1 \text{ClassLivestream}_{c,q} + \mathbf{X}'_{c,q-1}\Gamma + \varepsilon_{c,q}$$

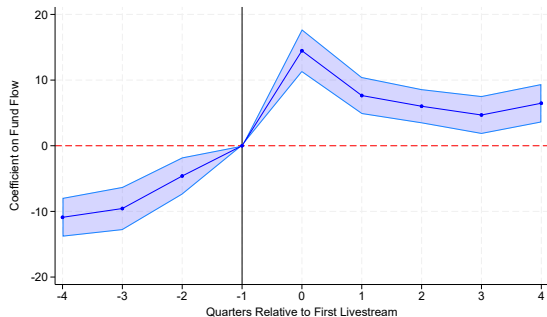
	(1)	(2)
Class Livestream	10.20*** (2.34)	
Solo Livestreamer (non-feat. class)		-9.82 (8.15)
Has Sibling Livestreamer		-19.91 (14.60)

Fund $\times$ quarter FE. SE clustered at family level.

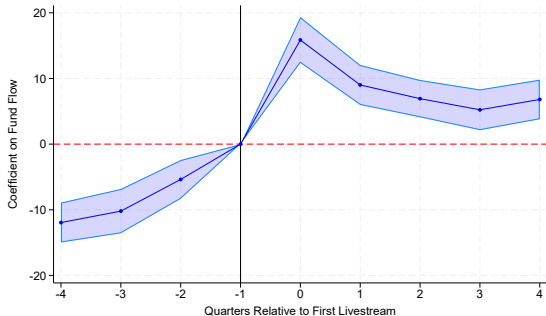
Featured classes gain **+10 pp** vs. non-featured classes of the *same fund* in the same quarter.
Reallocation operates at every level: family $\rightarrow$ fund $\rightarrow$ share class.

Appendix: Performance Event Study (Robustness)

TWFE



Family $\times$ Quarter FE



Both specifications confirm pre-event outperformance of featured funds and post-event convergence. Robust to controlling for all time-varying family shocks.

Appendix: LLM Scoring Dimensions (All 16)

1. Sentiment Positivity
2. **Past Performance Focus***
3. **Social Proof Reliance***
4. Emotional Intensity
5. Sales Pressure
6. Educational Content
7. Risk Transparency
8. Community Building
9. Urgency / FOMO
10. Product Complexity Framing
11. Presenter Credibility Signals
12. Interactive Engagement
13. Narrative / Storytelling
14. Regulatory Compliance Emphasis
15. Future Performance Claims
16. Comparative Benchmarking

* Significantly different between high- and low-abnormal-flow families ($p < 0.05$).

Scoring: Qwen-2.5-32B; 10,000-char overlapping segments; 1–10 scale; rubric fixed ex ante so measurement error is orthogonal to the strategies under study.